

ProjXChange – A Student-Investor Collaboration Platform.

Project Proposal

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● Introduction

ProjXChange is a two-sided platform designed to connect students and investors. It enables students to showcase their academic or self-initiated projects, seek guidance, and resolve technical issues through a collaborative community. Simultaneously, it provides investors with access to a pool of innovative projects for opportunities.

● Objective

To design and develop a collaborative platform where students can post projects and queries, and investors can bid to fund promising ideas, fostering innovation continuity and industry linkage.

● Problem Description

Student projects created during academic semesters often remain unused or abandoned due to a lack of continuation and exposure to industry support. On the other hand, investors face challenges in identifying raw but high-potential student innovations to fund. The absence of a unified digital space limits opportunities for collaboration, project reuse, and academic community support. ProjXChange solves this by enabling students to showcase ideas, receive multiple bids from investors. The platform also fosters a community where students can upload and resolve errors collaboratively, ensuring projects evolve and thrive.

● Target Industry

1. Education Technology (EdTech).
2. Startup & Investment Ecosystem.
3. Research & Innovation.
4. Community Platforms.

● Methodology

1. The system focuses into two modules (Student + Investor)
2. Front-end: React Native(Expo) for mobile apps, React.JS for admin dashboard.
3. Back-end: Node.IS with Express for API development.
4. Database: Neon or PostgreSQL for scalable storage.
5. Authentication: secure login/sign-up.
6. Payment Integration: Stripe API/easy paisa for secure transactions.
7. Development Approach: Agile methodology with iterative sprints and testing.

● **Project Scope**

The project will develop a student–investor platform where students can upload their projects and investors can browse and buy based on these projects. We will not cover advanced plagiarism detection, large-scale payment gateways, or legal copyright issues. It is assumed that both students and investors are genuine users, and the system will serve as an academic innovation prototype rather than a full-scale industry solution.

● **Feasibility Study**

With the above-defined scope, the proposed project is feasible within the given time frame of the Final Year Project schedule. The system is designed as a web-based platform, which can be developed using existing frameworks and tools that are readily available. The project’s functionality will initially focus on core modules (Student & Investor) to ensure timely completion, with optional features (advanced verification, AI checks) considered for future enhancements.

i. Risks Involved:

1. Project Authenticity Risk:

Risk: Students may upload copied projects from external sources.

Solution: In the prototype, projects will be assumed to be genuine. For future versions, a verification module or plagiarism check will be considered.

2. Time Management Risk:

Risk: Limited time to implement all advanced features.

Solution: Prioritize core features (upload, browse, buy, hire) first, and add optional features only if time permits.

3. Technical Complexity Risk:

Risk: Integration of payment gateways or large file uploads may be complex.

Solution: Use simple payment simulation and file upload system in the prototype.

4. User Engagement Risk:

Risk: Lack of real users (students/investors) during testing.

Solution: Use dummy data and sample projects for demonstration.

ii. Resource Requirement:

1. Computing Resources:

Internet connectivity for web hosting and API usage.

Software Tools:

1. Front-end: React Native (Expo) for mobile apps, React.JS for admin dashboard.
2. Back-end: Node.JS with Express for API development.
3. Database: Neon or PostgreSQL for scalable storage.
4. Authentication: secure login/sign-up.

5. Payment Integration: Stripe API for secure transactions.
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● **Solution Application Areas**

Our project creates value by connecting students and investors. Students can upload and showcase their projects, get recognition, funding, or job opportunities. Investors and companies can explore projects, purchase solutions, or hire talented students. Universities benefit by encouraging project re-usability and innovation. Overall, the platform reduces project wastage, promotes research, and supports entrepreneurship by turning student work into real-world impact.

● **Tools/Technology**

1. **Hardware Requirements:**

1. Standard PC/Laptop for development.
2. Internet connection

2. **Software Requirements:**

1. Front-end: React Native (Expo) for mobile apps, React.js for admin dashboard.
2. Back-end: Node.js with Express for API development.
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5. Payment Integration: Stripe API for secure transactions.
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● **Expertise of the Team Members**

Our team has the required knowledge from courses like Web Development, Databases, and Software Engineering, Mobile App Development, UI/UX. Members are skilled in frontend, backend. The project is of equal interest to all, and each member is motivated to contribute.

● **Milestones**

FYP-I (Semester 1)

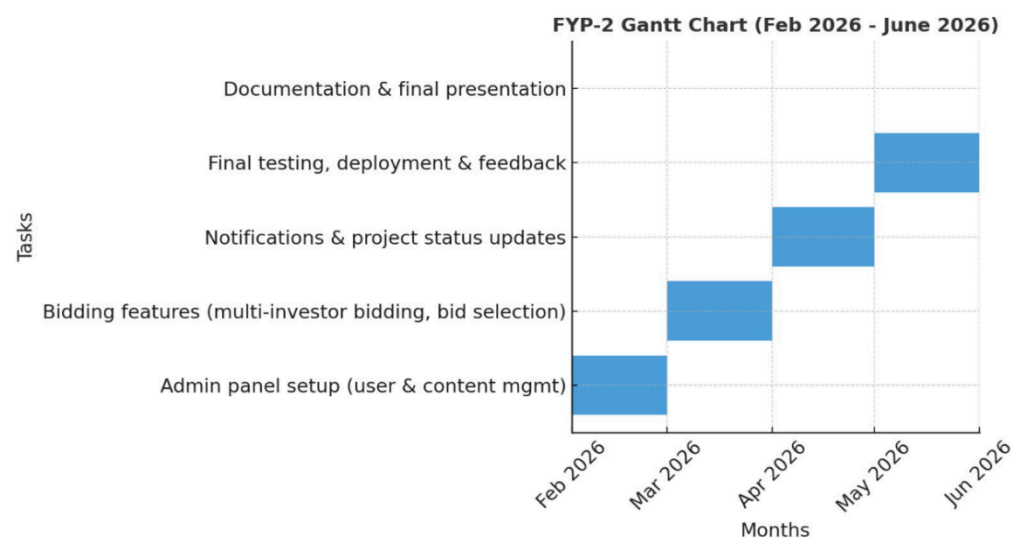
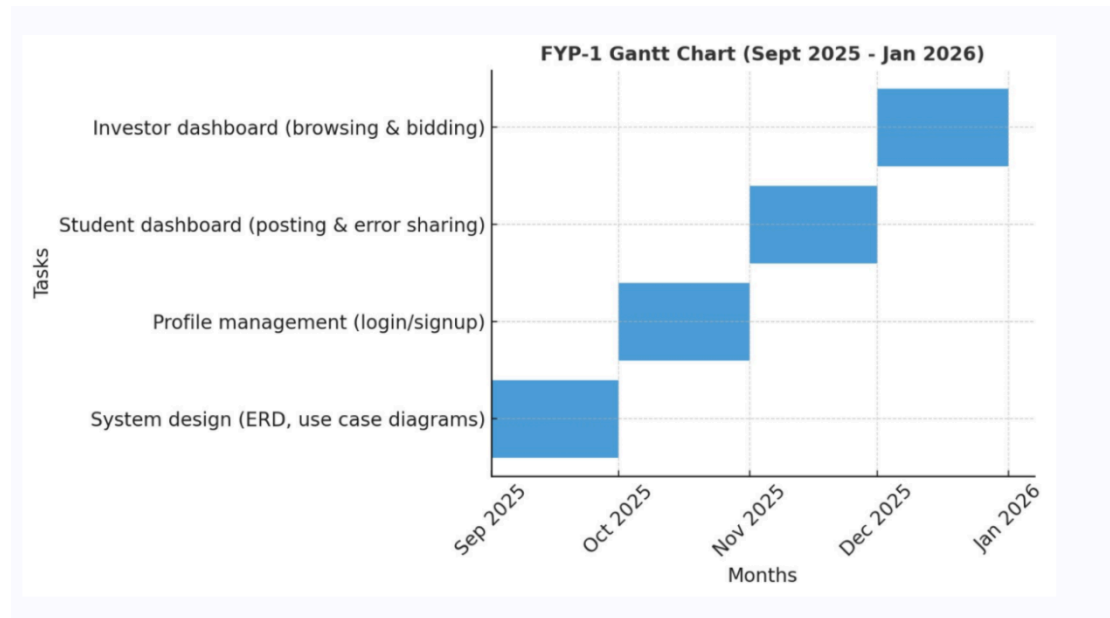
1. System design (ERD, use case diagrams).
2. User profile Management (signup/login).
3. Student dashboard (basic project posting and error sharing).
4. Investor dashboard (basic browsing and bidding).

FYP-II (Semester 2)

1. Admin panel setup (user and content management).
2. Bidding features (multi-investor bidding, bid selection).

3. Notifications and project status updates.
4. Final testing, deployment, and user feedback.
5. Documentation and final presentation.

● Project Schedule



● Work Breakdown Structure

FYP-1 (Fall Semester from Sept 2025 to Jan 2026)

Milestone: Basic system design completed and prototype dashboards functional.

Phase 1 – Requirements & Planning (Weeks 1–2)

Week 1 and 2:

- 1) Finalize project requirements and scope.
- 2) Decide SDLC methodology.

Phase 2 – System Design (Weeks 3–5)

Week 3:

- 3) Design ERD for database schema.
- 4) Draft Use Case and UML Diagrams for all user types.

Week 4:

- 5) Review/refine ERD and diagrams with feedback.
- 6) Document functional & non-functional requirements.

Week 5:

- 7) Finalize system design.

Milestone: System Design Approved.

Phase 3 – User Authentication

Week 6:

- 8) implement Signup/Login (backend + frontend).
- 9) Add basic validation and error handling.

Week 7:

- 10) Test and debug authentication.

Milestone: Secure Authentication Functional.

Phase 4 – User Profile Creation

Week 8:

- 11) Build profile creation for users UI.
- 12) Integrate backend for storing/updating user info.

Phase 5 – Dashboards Development

Weeks 9–10:

- 13) Develop Student Dashboard (project posting, error sharing).

14) Conduct unit testing.

Weeks 11–12:

15) Develop Investor Dashboard (project browsing, bidding features).

Week 13:

16) Integrate both dashboards.

17) Test interactions between modules.

Milestone: Prototype Dashboards Ready.

Phase 6 – Testing & Review

Week 14:

18) Perform integration testing across all modules.

Week 15:

19) Fix bugs and refine UI/UX.

Week 16:

20) Finally review the documentation, UI, Dashboards and other project things with our supervisor.

21) Prepare submission + final presentation.

FYP-2 (Spring Semester from Feb 2026 to June 2026)

Milestone: Fully functional platform deployed with bidding features and admin controls.

Phase 1 – Setup & Planning (Weeks 1–2)

Week 1:

- 1) Review FYP-1 feedback and finalize improvements.
- 2) Plan features.

Week 2:

- 3) Upgrade development environment.
- 4) Prepare updated database and structure.

Phase 2 – Admin Panel Setup (Weeks 3–5)

Weeks 3–4:

- 5) Develop Admin Panel UI (user & content management).
- 6) Implement backend CRUD operations.

Week 5:

- 7) Integrate Admin Panel with existing dashboards.

Milestone: Admin Panel Functional.

Phase 3 – Bidding Features (Weeks 6–8)

Weeks 6–7:

- 8) Build a Multi-Investor Bidding system.
- 9) Implement Bid Selection Logic.

Week 8:

- 10) Test and refine bidding workflows.

Milestone: Bidding features Ready.

Phase 4 – Notifications & Status Updates (Weeks 9–10)

Week 9:

- 11) Build notification system for project updates & bids.

Week 10:

- 12) Enable notifications & test flows.

Milestone: Notification System.

Phase 5 – Final Testing & Deployment (Weeks 11–14)

Weeks 11–12:

- 13) Perform end-to-end testing across all modules.

Week 13:

- 14) Fix bugs, optimize performance.

Week 14:

15) Deploy the final system to the production environment.

Milestone: System Deployed.

Phase 6 – Documentation & Presentation (Weeks 15–16)

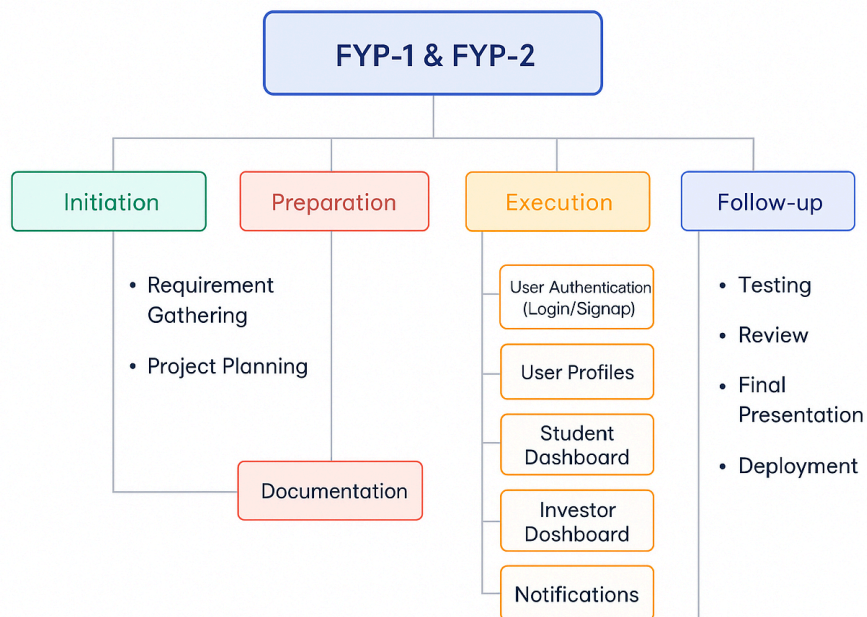
Week 15:

16) Prepare documentation & reports.

Week 16:

17) Conduct a final user feedback session and will show the final project details with our supervisor.

18) FYP-2 final presentation.



● **References**

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